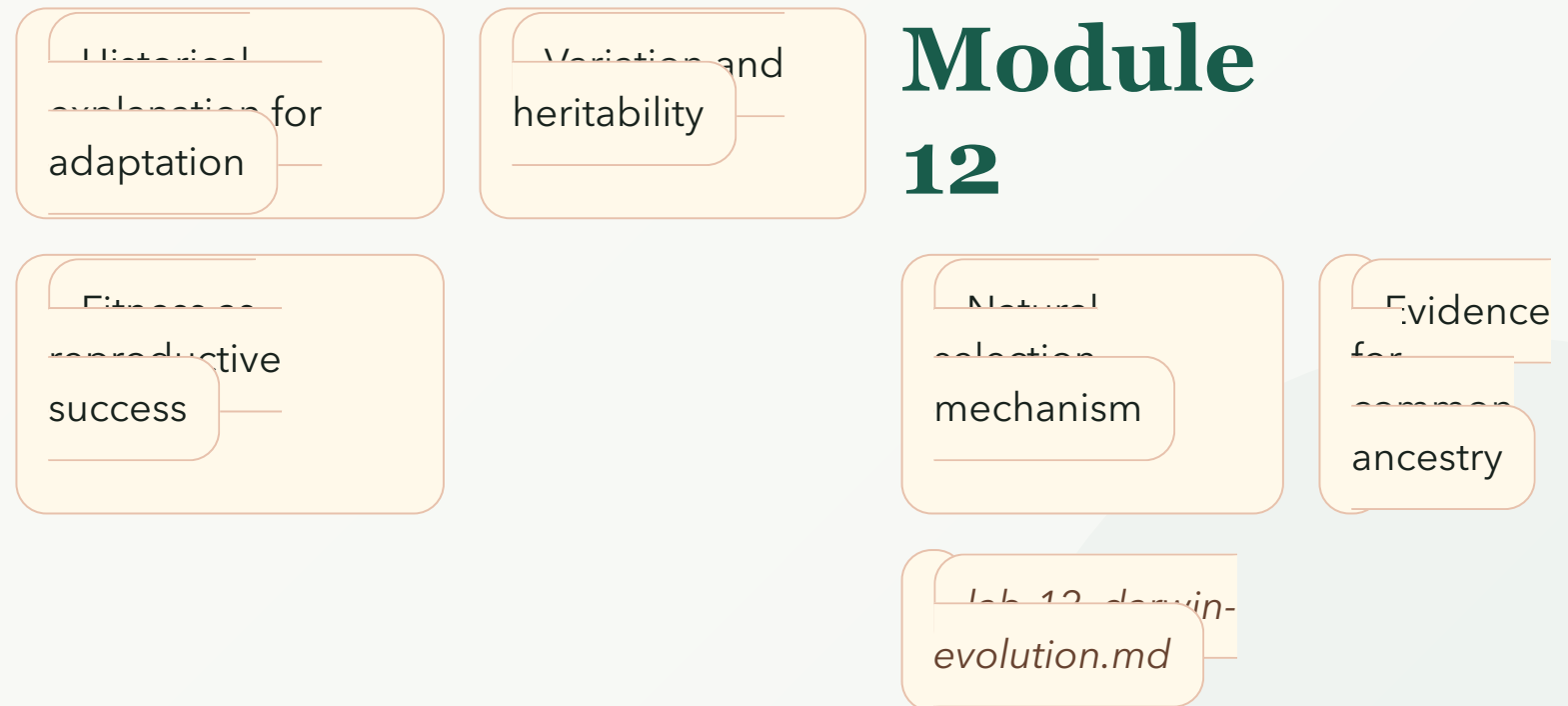


DARWIN AND EVOLUTION

Module map

- Module 12: Darwin and Evolution
- Connected lab: lab-12_darwin-evolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.



DARWIN AND EVOLUTION

Learning objectives

- Explain natural selection using variation, heritability, and reproductive success.
- Define fitness in evolutionary terms.
- Distinguish individual change from population evolution.
- Use multiple evidence types to support common ancestry.
- Apply natural selection to a concrete trait example.

OBJECTIVE 1

Explain natural selection using variation, heritability, and reproductive success.

OBJECTIVE 2

Define fitness in evolutionary terms.

OBJECTIVE 3

Distinguish individual change from population evolution.

OBJECTIVE 4

Use multiple evidence types to support common ancestry.

OBJECTIVE 5

Apply natural selection to a concrete trait example.

DARWIN AND EVOLUTION

Topic sequence

- Historical explanation for adaptation: Darwinian evolution explains adaptation through descent with modification.
- Variation and heritability: Natural selection requires variation, heritability, and unequal reproductive success.
- Natural selection mechanism: Fitness means relative reproductive success in a particular environment.
- Evidence for common ancestry: Fossils, homology, biogeography, and molecular evidence support common ancestry.
- Fitness as reproductive success: Evolution changes populations over generations, not individual organisms during life.

TOPIC 1

Historical explanation for adaptation

TOPIC 2

Variation and heritability

TOPIC 3

Natural selection mechanism

TOPIC 4

Evidence for common ancestry

TOPIC 5

Fitness as reproductive success

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Evidence Map

- Module focus: Darwin and Evolution
- Central claim: Darwin and Evolution explains natural selection from variation, heritability, fitness, and ancestry evidence.
- Key nodes to connect: Historical explanation for adaptation, Variation and heritability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-12_darwin-evolution.md supplies an evidence surface for the map.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Reasoning Sequence

- Module focus: Darwin and Evolution
- Trace the model: Historical explanation for adaptation -> Variation and heritability -> Natural selection mechanism -> Evidence for common ancestry
- Inputs become outputs: Historical explanation for adaptation, Variation and heritability -> Explain natural selection using variation, heritability, and reproductive success., Define fitness in evolutionary terms.
- Feedback check: lab-12_darwin-evolution.md evidence checks whether students can explain natural selection mechanism.

DARWIN AND EVOLUTION

Terms as evidence handles

- Evolution: Change in heritable traits of populations across generations.
- Natural selection: Nonrandom differences in survival or reproduction tied to heritable traits.
- Adaptation: Heritable trait shaped by selection in a context.
- Fitness: Relative reproductive success.
- Common ancestry: Shared origin of lineages through descent.
- Homology: Similarity due to shared ancestry.

EVOLUTION

Change in heritable traits of populations across generations.

NATURAL SELECTION

Nonrandom differences in survival or reproduction tied to heritable traits.

ADAPTATION

Heritable trait shaped by selection in a context.

FITNESS

Relative reproductive success.

COMMON ANCESTRY

Shared origin of lineages through descent.

HOMOLOGY

Similarity due to shared ancestry.

DARWIN AND EVOLUTION

Lab connection

- Lab file: lab-12_darwin-evolution.md
- Lab evidence should test or illustrate: Natural selection mechanism
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Historical explanation for adaptation

EVIDENCE

lab-12_darwin-evolution.md

CLAIM

Explain natural selection using variation, heritability, and reproductive success.

DARWIN AND EVOLUTION

Contrast check

- Do not stop at: Darwinian evolution explains adaptation through descent with modification.
- Stronger explanation: Evolution changes populations over generations, not individual organisms during life.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Darwinian evolution explains adaptation through descent with modification.

DEEPER

Evolution changes populations over generations, not individual organisms during life.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Retrieval and Lab Check

- Module focus: Darwin and Evolution
- Retrieval prompt: What are the required ingredients for natural selection?
- Answer check: Explain natural selection using variation, heritability, and reproductive success.
- Required terms: Evolution, Natural selection, Adaptation
- Lab connection: lab-12_darwin-evolution.md; lab-12_darwin-evolution.md supplies evidence for Darwinian evolution evidence from trait and environment comparisons.

DARWIN AND EVOLUTION

Practice quiz bridge

- 1. Which statement best defines Evolution in Module 12?
- 2. Which learning goal best supports the topic "Variation and heritability"?
- 3. How should lab-12_darwin-evolution.md support Module 12?
- 4. Which retrieval move best prepares a student for Darwin and Evolution?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

DARWIN AND EVOLUTION

Synthesis and exit ticket

- Exit claim: explain historical explanation for adaptation using evidence from lab-12_darwin-evolution.md.
- Must include: Evolution, Natural selection, Adaptation.
- Revision prompt: what evidence would change your explanation?

CLAIM

Historical explanation for adaptation

EVIDENCE

lab-12_darwin-evolution.md

REVISION

What would change your mind?